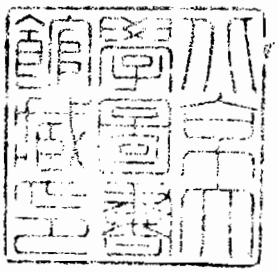
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The Nuclear Many-Body Problem

With 171 Figures





Springer

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Preface

It is the aim of this book to describe in concise form our present theoretical understanding of the nuclear many-body problem. The presentation of the enormous amount of material that has accumulated in this field over the last few decades may be divided into two broad categories: One can either concentrate on the physical phenomena, such as the single-particle excitations, rotations, vibrations, or large-amplitude collective motion, and treat each of them using a variety of theoretical methods; or one may stress the methodology and technical aspects of the different theories that have been used to describe the nucleus. We have chosen the second avenue. The structure of this book is thus dictated by the different methods used—Hartree–Fock theory, time-dependent Hartree–Fock theory, generator coordinates, boson expansions, etc.—rather than by the physical subjects.

Many of the present theories have, of course, already been presented in other textbooks. In order to be able to give a more rounded picture, we shall either briefly review such topics (as in the case of the liquid drop or the shell model) or try to give more updated versions (as in the cases of rotations or the random phase approximation). Our essential aim, however, is to present the more modern theories—such as boson expansions, generator coordinates, time-dependent Hartree–Fock, semiclassical theories, etc. —which have either never been seen, or at best had little detailed treatment in, book form.

The book is essentially directed towards students who have had a conventional course in quantum mechanics and have some basic under-

standing of nuclear phenomena. Our intention has been to close the gap between the usual graduate lecture course and the literature presented in scientific journals. We have therefore put as much emphasis as possible on clarity, and to this end often go into quite extensive mathematical detail. We hope that the reader will thus be able to rederive many of the formulas presented without too much difficulty.

Originally our idea was to make an updated translation of an existing book on Nuclear Models by G. Baumgärtner and P. Schuck [BS 68a]. However, we soon found that theoretical nuclear physics had evolved so rapidly over the last decade that a completely new book was called for. Nevertheless, the reader will find some remnants of the original book, for example, in Chapters 2 and 6. The editing of the present work has been undertaken by Springer-Verlag, to whom we are grateful for their very constructive collaboration.

We are indebted to many of our colleagues for innumerable discussions and helpful remarks. In the first place, we wish to express our thanks to R. R. Hilton for his most careful reading of the entire manuscript, and for pointing out along the way many conceptual and countless linguistic errors.

Specifically, we are particularly grateful to the following people for clarifying discussions on a number of topics: for the rotational problem in nuclear physics—R. Arvieu, L. Egido, R. M. Lieder, H. J. Mang, E. Marshalek, J. Meyer-ter-Vehn, J. O. Rasmussen, and F. S. Stephens; for the theory of finite Fermi liquids and its applications to collective nuclear excitations—V. Klemt, J. Speth, E. Werner, and W. Wild; for the description of anharmonicities in nuclear spectra using boson expansion techniques—S. T. Belyaev, G. Holzwarth, S. Iwasaki, E. Marshalek, T. Marumori, K. Matsuyanagi, R. Piepenbring, F. Sakata, M. Yamamura, and V. G. Zelevinskii; for the generator coordinate method—T. Fliessbach, K. Goeke, G. Holzwarth and P. G. Reinhard; for the time-dependent Hartree–Fock approach—K. Dietrich, J. J. Griffin, S. E. Koonin; and for semiclassical methods in nuclear physics—J. Bartel, R. Bengtsson, R. K. Bhaduri, M. Brack, M. Durand, H. Gräf, G. Holzwarth, and B. Jennings.

Thanks are also due to L. Egido for a careful reading of several chapters, to S. Iwasaki for a reading of the chapter on boson expansions, and to R. K. Bhaduri, M. Brack, M. Durand, and G. Holzwarth for a careful reading of the chapter on semiclassical methods.

Our nuclear physics education has taken place within the Munich group, and we are happy to be able to take this opportunity to thank W. Brenig, K. Dietrich, H. J. Mang, H. Schmidt, and W. Wild for the patience and understanding they have shown in the numerous discussions we have had over the years.

Many aspects of the book were clarified during several stays by one of us (P. R.) at the Institute Laue-Langevin in Grenoble, for whose support

we are most thankful. Both of us have also had occasion to discuss many subjects with our French colleagues at the Institut de Sciences Nucléaires, Grenoble; the Institut de Physique Nucléaire, Orsay; and the Centre d'Etudes Nucléaires, Saclay. We hope they will not mind if we thank them here in such a "pauschal" fashion.

In spite of all this help and collaboration it is almost inevitable in a book of this kind that some misconceptions will still remain; naturally, they are our own responsibility. However, suggestions and criticisms from our readers would be welcomed for some possible future edition.

Various hands have assisted in the task of typing the manuscript and we should like finally to thank our many secretaries for their valuable contribution.

Munich P. Ring
Grenoble P. Schuck
February 1980

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